Sahaj Bhadja

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Summary

Software engineer with 2+ years building **AI-powered full-stack products**, data pipelines, and production web apps. Strong in **Python**, **TypeScript**, **React/Next.js**, **SQL**, and shipping practical automation fast.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant platform services on Azure SQL and Blob Storage for 500+ users, optimizing query paths and data access patterns to reduce dashboard latency from 1.2s to 350ms in production.
- Built fault-tolerant async workflows for 1K+ AI jobs with explicit state transitions, failure isolation, and release-safe debugging paths, improving throughput 3x under concurrent production load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Flask services, REST APIs, and instructor-facing React workflows for AI grading operations handling 10K+ daily requests, integrating secure file ingestion, scoring pipelines, and production-ready release flows.
- Designed concurrent batch-processing for 1K+ uploaded files using multithreading and AWS RDS-backed services, maintaining stable API behavior while reducing grading turnaround time by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built spatial-temporal evaluation pipelines over 100K+ trajectory records, improving decision stability by 15% while strengthening reproducibility, measurement, and debugging across research-style AI workflows.
- Developed experimental tooling and reporting loops for sensor-driven systems, using iterative validation and failure analysis to communicate actionable findings clearly across interdisciplinary engineering collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered microservice-based software with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, improving release reliability through 20+ automated unit and integration test suites.
- Optimized API and ETL workflows with AWS Glue, S3, and SQL-backed services, improving response times by 30% and pipeline efficiency by 68% for production data operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed full-stack workflow software with REST APIs, WebSockets, and responsive interfaces, supporting 2M+ synchronized records while improving operational visibility and real-time coordination for users.
- Automated Docker-based build and deployment workflows with Jenkins CI/CD, cutting release time by 50% and improving troubleshooting speed, environment consistency, and team delivery reliability.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++, Bash
Frontend: React, Next.js, HTML5, CSS3, TailwindCSS, responsive UI, dashboard workflows
Backend & Data: Flask, FastAPI, Node.js, REST APIs, PostgreSQL, MySQL, Azure SQL, Supabase, ETL pipelines, Blob Storage, S3
Tools & AI: Git, Docker, Jenkins, CI/CD, Vercel, AWS, GCP, multithreading, OCR workflows, LLM integrations, prompt workflows

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Implemented consent-based patient-history access, appointment alerts, and operational dashboards with TypeScript-backed workflows, reducing administrative workload by 30% while improving reliability and issue visibility.

PiXMod Live | *Next.js, MongoDB, Stripe, CLIP, SAM, Stable Diffusion*
- Built an AI-powered image platform with Next.js, vision models, and scalable storage, supporting editing, search, and billing workflows while increasing premium conversion by 18%.

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, Kubernetes*
- Built a scalable fintech platform supporting 50K+ users and 1M+ transactions, combining secure APIs, real-time updates, and dashboard-style workflows under reliable CI/CD-backed deployments.